

## SAMEER KHAN

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Senior ML Engineer | Document AI • LLM • RAG • Agentic AI | AWS SageMaker • Databricks

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### PROFILE SUMMARY

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- Senior ML Engineer with **8 years** delivering production Document AI systems for financial services — **ICE Data Services** (Hyderabad) and **AI Rajhi Bank** (Riyadh, Saudi Arabia).
- Shipped a multimodal document classifier (CNN + BiLSTM + Transformer) achieving **94% page-level accuracy** on **100K+ mortgage documents**; built a FAISS vector search pipeline that reduced root-cause analysis time by **60%**.
- Specialised in LLM-powered document intelligence: **RAG pipelines** (FAISS + FastAPI + Streamlit), QLoRA fine-tuning (Mistral-7B), LangChain agents, and LangGraph agentic workflows.
- End-to-end pipeline ownership — multimodal extraction → distributed training (Horovod, - **30% training time**) → AWS SageMaker + Databricks deployment → MLflow monitoring.

### TECHNICAL SKILLS

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**LLM & GenAI** : RAG pipeline, LLM Fine-tuning (LoRA/QLoRA), Prompt Engineering, LangChain, LangGraph, FAISS, Sentence Transformers, HuggingFace  
**MLOPs** : SageMaker, Databricks, Jenkins CI/CD, Docker, MLflow, FastAPI, Evidently  
**NLP & DL** : BERT, Transformers, NER, BiLSTM, CNN, PyTorch, Scikit-learn  
**Cloud** : AWS (SageMaker, S3), Azure ML, Databricks, PySpark  
**Languages** : Python, SQL

### WORK EXPERIENCE

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#### Aug 2021 – Present | ICE Data Services, Hyderabad, India | Sr. ML Engineer

(Aug 2021-Nov 2023 via Persistent Systems; Nov 2023-Present direct hire)

- Architected a CNN + BiLSTM + Transformer document classifier for ICE's mortgage pipeline — **93% page-level / 85% document-level accuracy** across **100+ document types**; integrated BERT NER to push accuracy to **94%**, deployed on SageMaker.
- Deployed **Donut LLM** as an OCR-free document extraction system, enabling direct text retrieval from mortgage documents without traditional OCR — reducing preprocessing complexity and improving extraction reliability.
- Scaled a deduplication pipeline across **7.1M mortgage pages** — DonutSwin embeddings, FAISS IndexFlatIP cosine search, Union-Find clustering on **40 GPU nodes** (TorchDistributor + NCCL); cut training corpus by **67% pages and 69% documents**, eliminating template overfitting.
- Built a FAISS semantic search system for model root-cause analysis, replacing keyword lookup and cutting investigation time by **60%**.
- Owned MLOps across the document AI pipeline — Horovod distributed training (**-30% training time**), MLflow, Jenkins CI/CD, Databricks + SageMaker serving.

#### Aug 2019 – Mar 2021: Zyclyx (Client: AI Rajhi Bank, Riyadh, Saudi Arabia) | Data Scientist

- Designed a **ResNet50** document classification pipeline for **English-Arabic** banking documents — categorised incoming files by type to route each to the correct extraction workflow.
- Built Tesseract + TensorFlow OCR pipeline for key-field extraction from English-Arabic financial forms; developed **GRU-based NER** to identify customer, transaction, and compliance entities; integrated output with bank APIs for automated database validation.
- Built a **Signature Verification System** using deep learning to detect fraudulent and mismatched signatures in offline banking documents, reducing manual compliance review overhead.
- Deployed the full pipeline on-premise via Flask + Unicorn microservices — from raw Arabic document ingestion to structured API output for production banking workflows.

## Feb 2017 – Aug 2019: VISSINDIA, Hyderabad, India | Python & Computer Vision Engineer

- Built identity verification and multi-object tracking pipelines — face detection (Haar Cascade, HOG, CNN-based), feature matching (SIFT/SURF/FLANN), and TensorFlow object detection for real-time video tracking.
- Developed image preprocessing pipelines (morphological ops, denoising, thresholding, skew correction) to improve OCR accuracy — foundation carried into Document AI work at **AI Rajhi Bank** and **ICE Data Services**.

## PROJECTS

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### RAG System — Production (2026)

End-to-end RAG pipeline — sentence-transformers (all-MiniLM-L6-v2), FAISS IndexFlatIP, FastAPI backend (/ingest, /query, /evaluate), Streamlit UI. MRR=1.0 on ML domain evaluation set.

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### Document Deduplication Pipeline (ICE Data Services, 2023)

Distributed deduplication across **7.1M mortgage pages** — DonutSwin page embeddings, FAISS IndexFlatIP cosine search, Union-Find transitive clustering across **40 GPU nodes** (TorchDistributor + NCCL). Reduced training corpus by **67% pages** and **69% documents**, eliminating template overfitting.

### Signature Verification System (AI Rajhi Bank, 2020)

Deep learning pipeline to detect fraudulent and mismatched signatures in Arabic banking documents — contrastive learning model deployed on-premise via Flask + Gunicorn for production compliance workflows.

## CERTIFICATIONS & TRAINING

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Advanced Deep Learning Specialization - Coursera (CNNs, RNNs, LSTMs, Sequence Models, Optimization)  
Machine Learning Operations (MLOps) Specialization - Coursera (CI/CD, deployment, monitoring, ML pipelines)

## ACADEMIC DETAILS

**B.Tech. in Computer Science Engineering** - TKR Engineering College, JNTU, Hyderabad | 2016